

**Cambridge (CIE) IGCSE Physics**  
**Paper 4: Theory (Set F)**  
**Extended**

**Saturday 26 April 2025**

**Morning (Time: 1 hour 15 minutes)**

Total marks

**/ 80**

**Instructions**

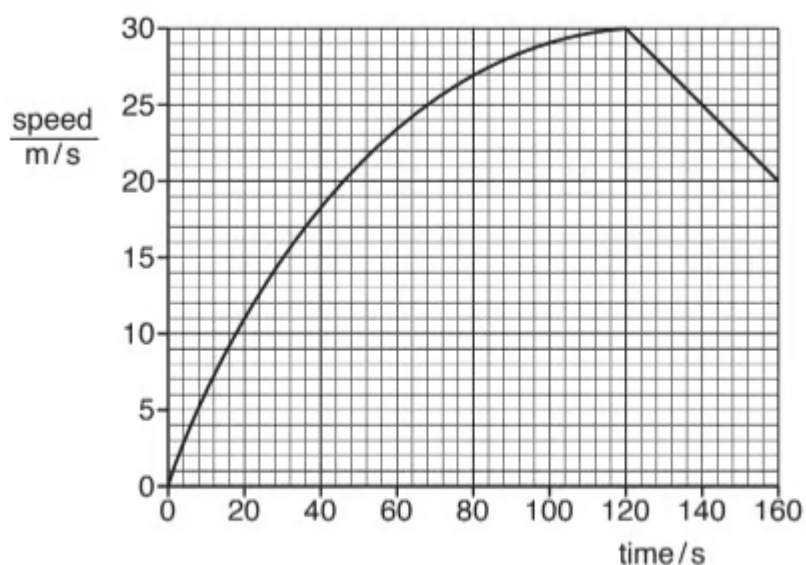
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall =  $9.8\text{m/s}^2$ ).
- You should show all your working and use appropriate units.

**Scan here to mark your mock exam**

or visit the mock exams landing page for this course



**1 (a)** Fig. 1.1 shows the speed-time graph for a vehicle accelerating from rest.



**Fig. 1.1**

Calculate the acceleration of the vehicle at time = 30s.

acceleration = .....

.....

.....

**(2 marks)**

**(b)** Without further calculation, state how the acceleration at time = 100 s compares to the acceleration at time = 10 s. Suggest, in terms of force, a reason why any change has taken place.

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.....

.....

**(3 marks)**

**(c)** Determine the distance travelled by the vehicle between time = 120 s and time = 160 s.

distance = .....

.....

.....

.....

**(3 marks)**

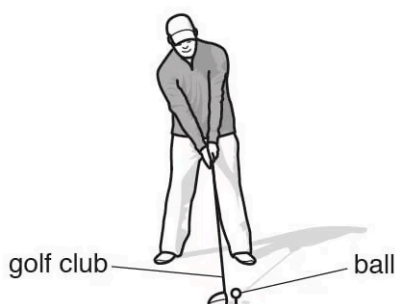
- 2 (a)** Complete Fig. 2.1 by writing in the right-hand column the name of the quantity given by the product in the left-hand column.

| product                    | quantity |
|----------------------------|----------|
| mass $\times$ acceleration |          |
| force $\times$ time        |          |

**Fig 2.1**

(2 marks)

- (b)** Fig. 2.2 shows a man hitting a ball with a golf club.



**Fig. 2.2**

The ball has a mass of 0.046 kg. The golf club is in contact with the ball for  $5.0 \times 10^{-4}$  s and the ball leaves the golf club at a speed of 65 m/s.

(i) Calculate:

1. the momentum of the ball as it leaves the golf club

momentum = .....[2]

2. the average resultant force acting on the ball while it is in contact with the golf club.

average force = .....[2]

(ii) While the golf club is in contact with the ball, the ball becomes compressed and changes shape.

Name the energy store being filled whilst the ball is in contact with the golf club.

[1]

.....

.....

.....

.....

.....

**(5 marks)**

**3 (a)** A solar panel receives energy from the Sun at a rate of 5.0 kW.

Thermal energy is transferred from the solar panel to water with an efficiency of 20%.

Cold water of mass 15 kg enters the solar panel every hour.

The specific heat capacity of water is  $4200 \text{ J / (kg}^\circ\text{C)}$ .

Calculate the temperature increase of the water.

temperature increase = .....  $^\circ\text{C}$

.....

.....

.....

.....

**(4 marks)**

**(b)** State and explain one advantage and one disadvantage of heating the water in a solar panel compared with heating the water in a coal-burning boiler.

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.....

.....

.....

**(4 marks)**

**4 (a)** Fig. 4.1 represents an atom.



**Fig. 4.1**

Representing atoms by circles approximately the same size as in Fig. 4.1, sketch

(i) on Fig. 4.2, the arrangement of atoms in a crystalline solid,

[1]

(ii) on Fig. 4.3, the arrangement of atoms in a gas.

[1]



**solid**

**Fig. 4.2**



**gas**

**Fig. 4.3**

.....  
.....  
**(2 marks)**

**(b)** (i) Describe the motion of the atoms in a solid.

[1]

(ii) A sculptor makes a statue from a block of crystalline rock using a cutting tool. Explain why he must apply a large force to the tool to remove a small piece of rock.

[2]

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**(3 marks)**

**(c)** A helium-filled balloon in the room of a house suddenly bursts.

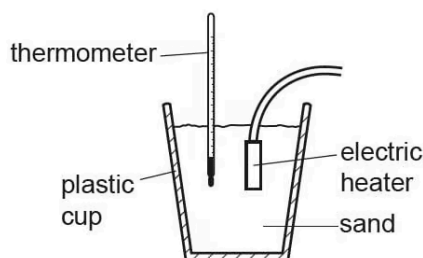
State and explain, in terms of atoms, what happens to the helium from the balloon after the balloon has burst.

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**(2 marks)**

- 5 (a)** Fig. 5.1 shows a plastic cup. The cup contains sand, an electric heater and a thermometer.



**Fig. 5.1**

The power of the heater is 50 W. The mass of the sand in the cup is 550 g. The initial temperature of the sand is 20 °C. The heater is switched on for 2.0 minutes.

The temperature is recorded until the temperature stops increasing. The highest temperature recorded by the thermometer is 33 °C.

- (i) Calculate the energy supplied by the heater.

energy = ..... [2]

- (ii) Calculate a value for the specific heat capacity of the sand, using your answer to (a)(i) and the data in the question.

specific heat capacity = ..... [3]

(iii) Explain why the specific heat capacity of sand may be different from the value calculated in **(a)(ii)**.

[2]

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**(7 marks)**

**(b)** On a sunny day, the temperature of the sand on a beach is much higher than the temperature of the sea.

Explain why.

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**(2 marks)**

- 6 (a)** A vibrating source on a ship produces a sound wave that travels through the ocean. The wave produced is a longitudinal wave.

Explain what is meant by the term longitudinal wave.

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**(3 marks)**

- (b)** The frequency of the sound wave is 800Hz.

- (i) The speed of sound in air is 330m/s.

State a typical value for the speed of sound in a liquid.

[1]

- (ii) Using your value from **(b)(i)**, calculate the wavelength of the sound wave in the ocean.

wavelength = .....[2]

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**(3 marks)**

7 (a) Fig. 8.1 shows a circuit.

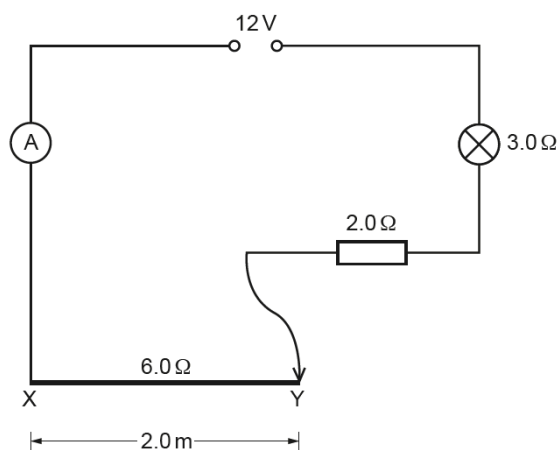


Fig. 8.1

The lamp has a resistance of  $3.0\ \Omega$ . Line XY represents a uniform resistance wire of resistance  $6.0\ \Omega$ .

Calculate the reading on the ammeter.

ammeter reading = .....

.....

.....

.....

(3 marks)

(b) Fig. 8.2 shows the circuit with a different connection to the resistance wire and an added resistor.

The length XY of the whole resistance wire is 2.0 m. The contact is made at Q where the distance XQ is 0.60 m.

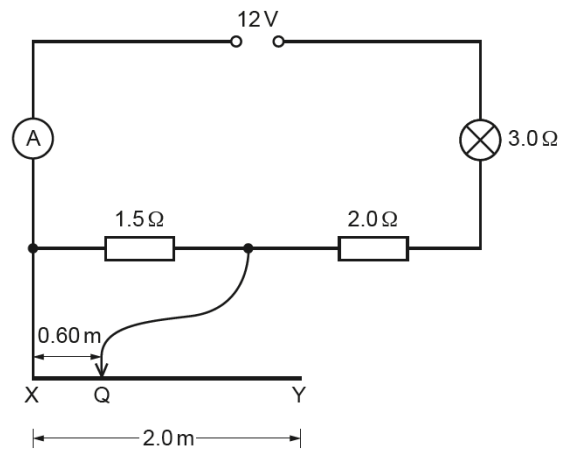


Fig. 8.2

Calculate the resistance of the circuit.

resistance = .....

.....

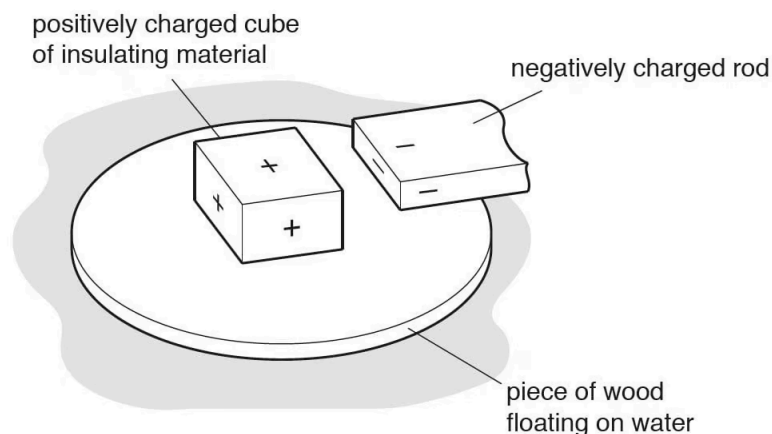
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(4 marks)

- 8 (a)** (i) Fig. 8.1 shows a positively charged cube of insulating material. The cube is fixed to a piece of wood that is floating on water. A negatively charged rod is held above the piece of wood and brought close to the cube, as shown.

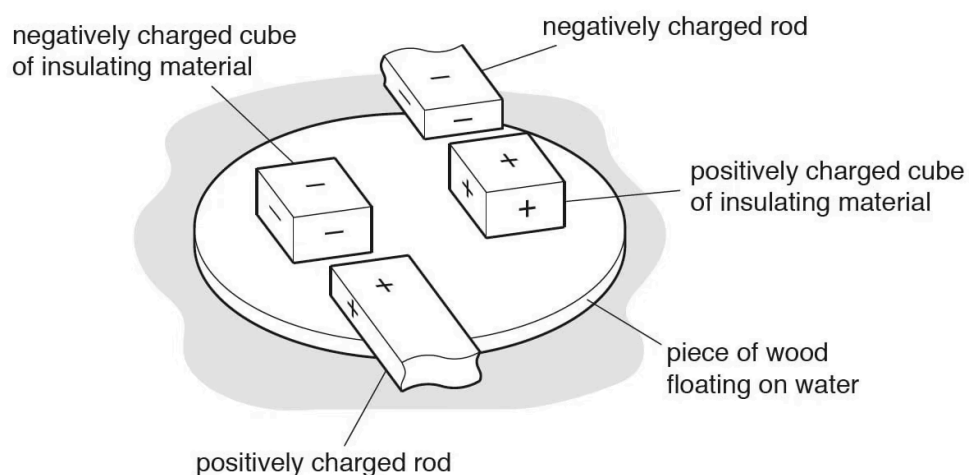


**Fig. 8.1**

State and explain any movement of the piece of wood.

[2]

- (ii) Fig. 8.2 shows two cubes of insulating material. One is positively charged and the other is negatively charged. The cubes are fixed to a piece of wood that is floating on water. Charged rods are held above the piece of wood and brought close to the cubes, as shown.



**Fig. 8.2**

State and explain any movement of the piece of wood.

[2]

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**(4 marks)**

**(b)** In terms of a simple electron model, describe the differences between conductors and insulators.

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**(2 marks)**

(c) On Fig. 8.3, draw the electric field pattern around a single-point positive charge.



**Fig. 8.3**

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**(1 mark)**

**9 (a)** State the type of radioactive emission that causes

(i) the proton number of a nuclide to increase by 1,

[1]

(ii) the nucleon number of a nuclide to decrease by 4,

[1]

(iii) no change in the proton number and no change in the nucleon number of a nuclide.

[1]

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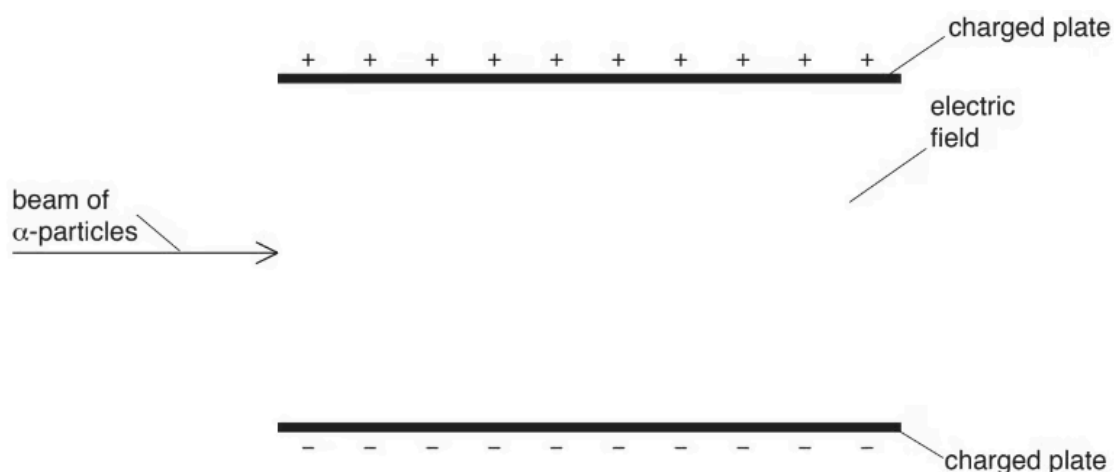
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**(3 marks)**

**(b)** The isotope radon-220 is radioactive and it decays by  $\alpha$ -particle emission.

(i) Fig. 11.1 shows a beam of  $\alpha$ -particles entering the electric field between two charged plates.



**Fig. 11.1**

On Fig. 11.1, sketch the path that the beam of  $\alpha$ -particles follows in the electric field.

[1]

(ii) The half-life of radon-220 is 56 s.

A sample of this isotope contains  $7.2 \times 10^6$  atoms.

Predict the number of  $\alpha$ -particles that the radon-220 in the sample emits in the next 168 s.

number of  $\alpha$ -particles emitted = .....[3]

.....

.....

.....

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(4 marks)

**10 (a)** Table 1.1 shows data about nine elements.

**Table 1.1**

| proton number | element   | symbol |
|---------------|-----------|--------|
| 2             | helium    | He     |
| 3             | lithium   | Li     |
| 4             | beryllium | Be     |
| 5             | boron     | B      |
| 6             | carbon    | C      |
| 7             | nitrogen  | N      |
| 8             | oxygen    | O      |
| 9             | fluorine  | F      |
| 10            | neon      | Ne     |

Carbon-14 is a radioactive isotope with a nucleon number of 14. It decays by emitting  $\beta$ -particles.

Use data from Table 1.1 to write down the nuclide equation for this decay.

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(4 marks)

- (b) A radioactive sample is placed close to a detector. The radioactive isotope in the sample has a long half-life. The detector records a count rate of 597 counts/s. Fig. 11.2 shows the readings when different materials are placed between the radioactive sample and the detector.

| Material                  | $\frac{\text{count rate}}{\text{counts/s}}$ |
|---------------------------|---------------------------------------------|
| a sheet of paper          | 602                                         |
| a piece of thin aluminium | 598                                         |
| a piece of thin lead      | 510                                         |

Fig. 11.2

Explain whether any  $\alpha$ -particles,  $\beta$ -particles or  $\gamma$ -rays are emitted by the radioactive sample.

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(3 marks)

**11 (a)** Some main sequence stars become black holes.

Describe the evolution of a main sequence star to the point at which it becomes a black hole.

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**(5 marks)**

**(b)** State and explain whether the Sun will eventually become a black hole.

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**(2 marks)**